Predictive Thermal Battery Management System for Li-ion Battery Packs

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Abstract

An internal short inside one cell is often named as the cause of lithium-ion battery accidents. In its most common form it starts small: it slowly drains its own cell and makes only a fraction of a watt of heat. A normal battery management system (BMS) cannot see this stage, because it acts only when a voltage, current or temperature crosses a fixed limit. I am building a four-cell pack (4S1P, NCR18650GA) with two layers of protection: an independent hardware BMS, and an ESP32 supervisor that predicts. Five times a second, the ESP32 runs a physical thermal model of every cell with a Kalman filter. It estimates what no sensor can measure: the core temperature, and the abnormal heat, in watts, that the cell's own current cannot explain. It forecasts each cell's temperature 1 and 5 minutes ahead and checks every forecast against the real reading. At rest, it estimates each cell's self-discharge from its voltage. All the evidence becomes one risk level (Normal, Watch, Warning or Trip) that can open separate charge and load relays and latch them.

Benchmark against existing solutions

Solution	What it acts on	Earliest warning	Needs before use	Disconnects
Hardware BMS board (SLCT 4S, ₹239; Layer A of my pack)	Fixed voltage and current limits; no temperature function	None	Nothing	Yes
Protection chips (TI BQ76952, NXP MC33772C)	Fixed temperature limits, e.g. 55 °C charging	None found in their datasheets	Settings	BQ76952 only
Orion Jr 2 BMS (US$405)	A current limit for every 5 °C	None	Settings	Via a contactor
Li-ion Tamer gas sensor (US$596)	Electrolyte vapour, after a cell vents	6–17 min before runaway (Sandia tests)	Nothing	No, signal only
Chen 2024; Sun 2022 (research)	Temperature patterns; outlier cells in a fleet	23–26 min; 660 s before runaway	Data from cells or a car taken into runaway	No
Ojo 2021 (base paper)	Neural-network temperature gap above 0.69 °C	Heating above about 9 °C/min	Training for each cell type	No
My project (supervisor parts ₹1,263)	Abnormal heat (W), 5-minute forecast check, leak current	0.4 W of abnormal heat (0.2 W over 5 min), or a 20 mA leak at rest	11 constants from normal use	Yes, two latched relays

Delta X over the base paper (Ojo et al., IEEE TIE, 2021)

	Base paper (measured)	My project (design target)
Healthy-cell prediction	Neural network, about 21,000 trained weights; retrained for each cell type and as the cell ages	Physical model and Kalman filter: 11 constants with units, fitted once; no training data
Slowest heating caught	About 9 °C/min: a 0.69 °C gap inside its 4.5 s window	Warning at 0.5 °C/min (0.4 W): **18×** slower heating; the 5-minute check catches 0.2 W: **34×**
Evidence used	Surface temperature gap only	Core temperature, abnormal heat, leak current (from 20 mA), resistance; thermal and electrical checked by an energy balance
Mistaken alarms	0.5 % of samples: up to 36 an hour at 2 Hz	Each level held 10–30 s; a soft fault needs thermal and electrical evidence to agree
Hardware	Jetson TX2 or Raspberry Pi; 45 ms per estimate on its CPU	₹639 ESP32; about **2,000×** less arithmetic per estimate; under 1 % of the processor
Scale and action	One cell on a lab tester; flags only	Four-cell pack, each cell compared with the others; opens latched relays, over an independent BMS

Worked example: a soft internal short

Gate 11 copies such a short safely, with a 47 Ω resistor bonded across one cell at rest (about 79 mA and 0.29 W; no fault inside the cell). With the estimated constants, the cell heats at only 0.39 °C/min and settles near 34 °C on a 25 °C day, so the hardware BMS, a 55 °C limit and a raw rate-of-rise warning all miss it. The base paper would see a gap of 0.03 °C against its 0.69 °C threshold. My design sees the cell 1.7 °C hotter than its 5-minute forecast (thermal Watch) and a leak four times the 20 mA level (electrical Watch); both agree on the same cell, and the level becomes **Warning**.

Two semesters

Semester 1 (Gates 0–2, no lithium cells): design, ESP-IDF firmware in C++, calibration of all 10 channels, relay fail-safe tests, and the first version of the model, filter, forecasts and risk levels, tested on synthetic data and a heated dummy. **Semester 2 (Gates 3–12):** cell matching and the welded pack, fitting the model, the safety state machine, Gate 11 validation, the Wi-Fi dashboard and the enclosure.

Within the papers reviewed, none combines abnormal heat in watts, confirmed forecasts, an electrical cross-check and a safe-current forecast on a low-cost controller. My figures are design targets (architecture revision 2.2), to be proven at Gates 9–11; no results are claimed yet. Ratios are my own calculations from the papers' figures, shown in full in the Benchmarking and Delta X document.